22AIM821- AI ETHICS FOR AIML ENGINEERS

AI ETHICS FOR AIML ENGINEERS

Course Code	**22AIM821**	**CIE Marks**	**50**
L:T:P:S	**3:0:0:0**	**SEE Marks**	**50**
Hrs/Week	**3**	**Total Marks**	**100**
Credits	**03**	**Exam Hours**	**03**

Course out comes: At the end of the course, the student will be able to:

22AIM821.1	Understand the legal and ethical frame works governing artificial intelligence.
22AIM821.2	Apply human rights-centered design, deliberation, and normative modes to mitigate ethics and address conflicts
22AIM821.3	Examine the oral framework of justice in AI and accountability in computer systems.
22AIM821.4	Evaluate the ethical implications of AI in health, public, legal fields.
22AIM821.5	Develop the ethical considerations of AI and its impact on society.
22AIM821.6	Collaborate with experts from various domains to build a cohesive ethical AI systems.

Mapping of Course Outcomes to Program Outcomes and Program Specific Outcomes:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
22AIM821.1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
22AIM821.2	3	-	-	-	-	-	-	-	-	-	-	2	2	-
22AIM821.3	-	3	-	-	-	-	-	-	-	-	-	2	2	-
22AIM821.4	-	-	-	3	2	-	-	2	-	-	-	2	2	-
22AIM821.5	-	-	3	-	-	-	-	3	-	-	-	2	2	2
22AIM821.6	-	-	-	-	-	3	-	3	-	2	-	3	2	3

MODULE-1	**Introduction and Overview**	21AIM814.1	**8Hours**
Introduction & Overview for Law and Regulation, Ethics of the Ethics of AI, Ethical Issues in Relationship with Artificial Entities.			
Text Book	Text Book 1: Ch 1		
MODULE-2	**Framework and Modes**	21AIM814.2	**8 Hours**
AI Governance by Human Rights- Centered Design, Deliberation and Oversight: End to Ethics Washing, The Incompatible Incentives of Private-Sector AI. Normative Modes: Codes and standards. The Role of Professional Norms in the Governance of Artificial Intelligence.			
Text Book	Text Book 1: 4-7		
MODULE-3	**Concepts and Issues**	21AIM814.3, 21AIM814.4	**8 Hours**
Moral Framework of Justice in AI: on the Limits, Failing and Ethics of Fairness, Accountability in Computer Systems-Responsibility and AI, The concept of Handoff as a Model for Ethical Analysis and Design.			
Text Book	Text book 1: 8 – 21		
MODULE-4	**Perspectives and Approaches**	21AIM814.4	**8 Hours**
Perspective on Ethics of AI - Computer Science, Social Failure Modes in Technology and the Ethics of AI: An Engineering Perspective, Automating Origination: Perspectives from the Humanities, Perspectives on Ethics of AI: Philosophy			
Text Book	Text Book 1: Ch 22- 28		
MODULE-5	**Cases and Application**	21AIM814.4, 21AIM814.5, 21AIM814.6	**8 Hours**
Ethics of AI in Transport - Ethics of AI in Biomedical Research, Patient Care and Public Health, Ethics of AI in Law: Basics Questions, Beyond Bias:" Ethical AI" in Criminal Law.			
Case Study	Solar Lighting Example		
Text Book	Text book 1: Ch 35-39.		

Suggested Learning Resources:

Text Books:

1. Mark us D Dubber, Frank Pasquale, Sunit Das," The Oxford Handbook of Ethics of AI", Oxford Press,2020. ISBN: 978-0-19-006739-7.

Reference Books:

1. Melanie Mitchell, "Artificial Intelligence: A Guide for Thinking Humans" Farrar, Straus and Giroux,2019. ISBN:978-0374257835
2. Patrick Lin, Keith Abney, and Ryan Jenkins, "RobotEthics2.0: From Autonomous Cars to Artificial Intelligence", OUP USA,2017. ISBN:978-0190652951.

Web links and Video Lectures(e-Resources):

- https://ocw.mit.edu/courses/res-ec-001-exploring-fairness-in-machine-learning-for-international-development-spring-2020/pages/module-one-introduction/

Activity-Based Learning (Suggested Activities in Class)/Practical-Based Learning

- Group discussion on real-world problems.
- Seminars

Introduction to AI Ethics

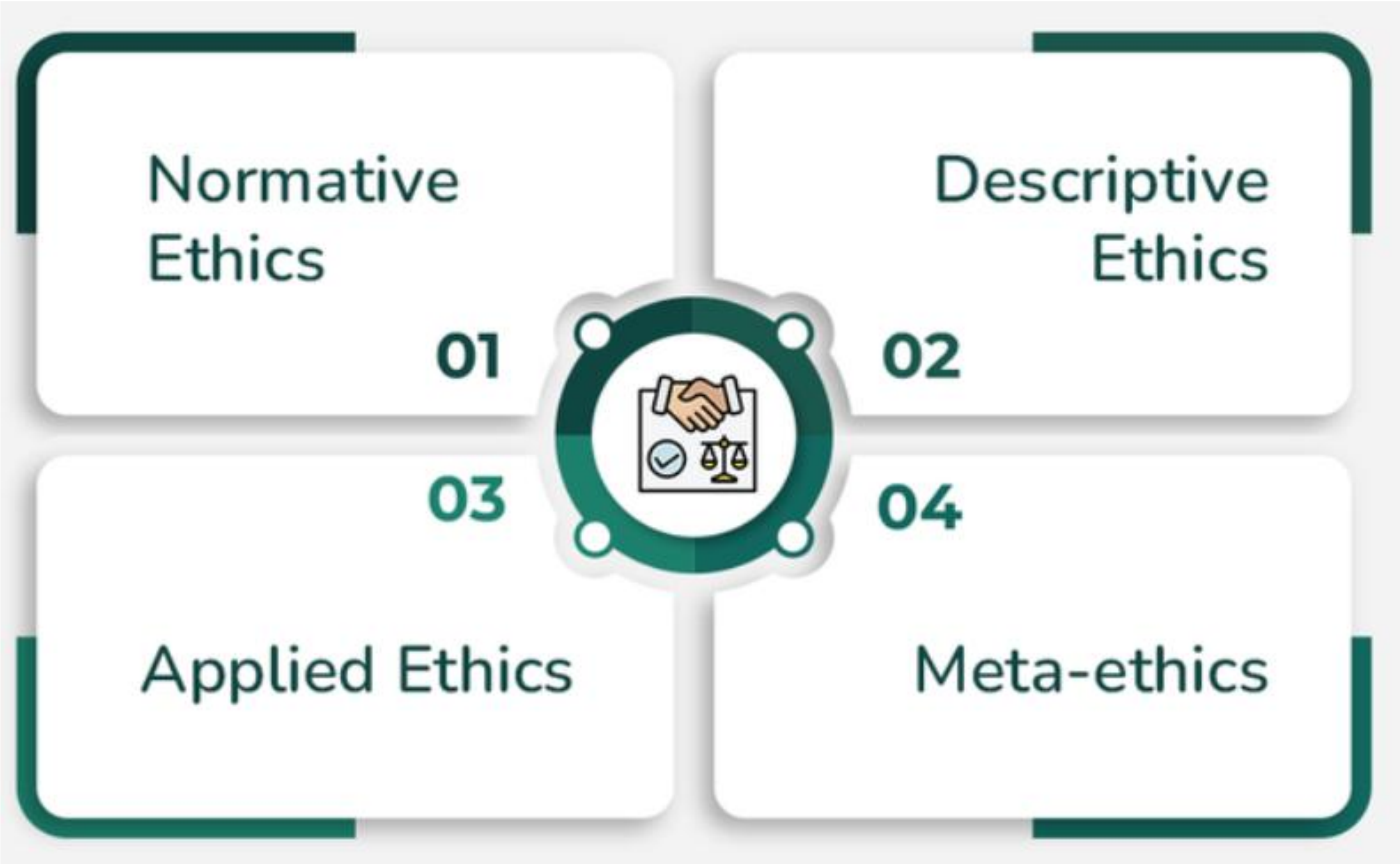
- Ethics deals with moral principles that guide human behavior.
- AI ethics applies moral principles to intelligence system
- It ensures AI systems are developed and used responsibly.
- AI ethics makes sure that:
  - AI is **fair** (it does not treat people unequally)
  - AI is **safe** (it does not cause harm)
  - AI **respects privacy** (it protects personal information)
  - AI is **responsible** (humans are accountable for its actions)

  Example:
  If an AI is used to select students or employees, it should choose based on ability, not on gender, caste, or background.
- AI ethics helps distinguish between acceptable and unacceptable AI behavior.

Why AI Ethics is Important

- AI systems influence critical decisions (loans, hiring, healthcare, policing).
- Unethical AI can cause bias, discrimination, and social harm.
- Lack of ethics can reduce trust in AI technologies.
- Ethical AI promotes public confidence and social acceptance.

Types of Ethics



Normative –

- Normative ethics deals with what should be done and how people ought to behave.
- It focuses on deciding what is right and what is wrong.
- It provides rules, principles, or guidelines that help people choose the correct actions in different situations.
- Examples of Normative Ethics
- Honesty – People should tell the truth and not lie.
- Fairness – Everyone should be treated equally without bias.
- Justice – Decisions should be made in a fair and lawful way.

Descriptive Ethics

- Studies how people **actually behave**
- Describes moral beliefs and practices of individuals or societies
- Does not judge right or wrong
- Based on observation and research

Example: Describes *how ethics is practiced in real life*

Applied Ethics

- Applies ethical principles to real-world situations
- Deals with practical issues

Examples:

- Medical ethics
- Business ethics

- **Meta Ethics**
- **Meta ethics** is concerned with the **meaning of ethical concepts and terms**.
  Instead of telling us **what actions are right or wrong**, it asks **what ethical words actually mean**.
- It studies questions such as:
- What does **“good”** really mean?
- What does **“fair”** mean?
- In AI, what does **“fair AI”** actually mean?
- Example :
- When we say **“Lying is wrong”**, Meta Ethics asks:
- What does **“wrong”** mean?
- Is it a fact, or just a belief?
- When we say **“Helping others is good”**, Meta Ethics asks:
- What does **“good”** mean?
- Is “good” the same for everyone?

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What is AI Ethics

- AI Ethics is the branch of ethics that deals with the moral principles and guidelines governing the design, development, deployment,and use of Artificial Intelligence (AI) systems.
- Key Principles of Ethical AI

- **Fairness**
  - AI systems should not discriminate. For example, an online exam evaluation system should not reduce marks based on a student's background.
- **Transparency**
  - AI decisions should be explainable. For example, if a loan is rejected, the applicant should know why.
- **Accountability**
  - Someone must be responsible for AI decisions. For example, if an autonomous car causes an accident, the company cannot blame only the AI. Someone must be responsible for AI actions
- **Privacy**
  - User data must be protected. For example, fitness apps should not share health data without permission.
- **Safety**
  - AI should not cause harm
- Without these principles, AI becomes dangerous.

Real Ethical Problems

- **Amazon Hiring Bias**
  - AI trained on past data
  - Favored male candidates
  - Showed gender bias
- **Google Photos Mislabeling**
  - Incorrect image labeling
  - Caused racial offense
- **Deepfake Scams**
  - Fake videos for fraud
  - Political misinformation
- **Autopilot Accidents**
  - AI-driven cars caused accidents
  - Who is responsible? AI or human?

These incidents clearly show why AI ethics is required.

Applications

- Healthcare Decision Support:
- Autonomous Vehicles
- Financial Loan Approvals
- Hiring Automation
- Smart Surveillance

Applications

- AI ethics is applied in many areas.
- **Healthcare**
  - AI assists doctors
  - Ethics ensures patient safety
- **Autonomous Vehicles**
  - AI makes driving decisions
  - Ethics guides life-critical choices
- **Financial Systems**
  - Loan approvals
  - Prevent unfair rejection
- **Hiring Automation**
  - Avoid biased recruitment
- **Smart Surveillance**
  - Balance security and privacy

Advantages

- **Benefits of AI Ethics**
- **Trustworthy Systems**
  AI Ethics helps build **transparent, reliable, and accountable** AI systems, increasing user trust.
- **Reduced Legal Risks**
  Ethical AI follows laws and regulations, reducing the risk of **legal issues, penalties, and lawsuits**.
- **Fair Treatment**
  AI Ethics ensures **equal and unbiased treatment** of all individuals, preventing discrimination.
- **Long-Term Sustainability**
  Ethical AI promotes **responsible innovation**, ensuring AI systems remain beneficial and acceptable to society in the long run.

LIMITATIONS

- Lack of Clear Definitions
- Bias in Training Data
- Black-Box Nature of AI
- Difficulty in Accountability
- Economic and Commercial Pressure

- **Challenges in AI Ethics**
- **Lack of Clear Definitions**
  There is no universal agreement on what is considered *ethical AI*, making it difficult to set common standards and rules.
- **Bias in Training Data**
  AI systems learn from historical data, which may contain social, gender, or racial biases. This can lead to **unfair and discriminatory outcomes**.
- **Black-Box Nature of AI**
  Many AI models work like a *black box*, where decision-making processes are not easily understandable, reducing transparency and trust.
- **Difficulty in Accountability**
  When AI systems cause harm, it is unclear **who should be held responsible**—developers, organizations, or users.
- **Economic and Commercial Pressure**
  Companies often prioritize **speed, profit, and competition**, which can lead to neglecting ethical considerations.

Introduction to Laws and Regulation in AI

- **What do we mean by Laws and Regulation**
  – Law is a rule made by the government
  – Everyone must follow the law

Breaking the law leads to punishment

  – Fine
  – Jail

- **Laws exist to maintain:**

1.Order
Laws help keep discipline and control chaos.
2.Safety
Laws protect people from harm, crime, and misuse of power.
3.Fairness in society
Laws ensure that everyone is treated equally and justly.

Why Do We Need Laws at All?

- Imagine a Society Without Laws
  - Anyone can do anything
  - Strong people can exploit weak people
  - No safety
  - No fairness
  - No justice

- Life in such a society would become **chaotic, dangerous, and unfair**.
- That is why **laws are necessary**.
- Laws:
  - Control human behavior
  - Protect the weak
  - Maintain safety
  - Ensure fairness and justice

- Now Think in Technical Terms
  - Any algorithm can behave in any way
  - Powerful AI systems can misuse data
  - Weak users have no protection
  - Outputs may be biased or harmful
  - Uncontrolled AI = Uncontrolled power

Why AI Needs Laws and Regulation

- Artificial Intelligence is **not normal software**.
- AI systems:
  - Learn from large datasets
  - Make decisions automatically
  - Affect **many people at the same time**
  - Because of this, **AI has power**.
  - Whenever something has power, it must be **controlled**.
- When AI systems operate without:Laws,Regulations,Ethical guidelines
  - They can cause harm at a **large scale**, faster than humans.
- That is why **AI needs laws, regulations, and ethical principles**,just like society needs laws to function safely and fairly."

Simple Real-Life Comparison

- Electricity is powerful → We have **safety rules**
- Vehicles are powerful → We have **traffic laws**
- Similarly:
- AI is powerful
  - AI can affect human lives
  - That is why **AI needs laws and regulations**

What Can Go Wrong If AI Has No Laws?

- If AI is used without laws and regulation:
  - AI can invade personal privacy
  - AI can discriminate between people
  - AI can make unfair decisions
  - AI can cause physical or mental harm
  - No one can be held responsible
  - Uncontrolled AI = Risk to society

Simple Example

- AI rejects your job application
- AI rejects your loan request
- No explanation is given
- Without laws:
  - You cannot question the system
  - You cannot complain to anyone
  - Laws exist to protect humans from AI misuse

Law vs Ethics vs Regulation

- **Ethics** → Moral values
- **Law** → Legal rules
- **Regulation** → How to follow the law

Ethics

- **Ethics** are:
  - **Moral principles** of right and wrong
  - Based on **human values**
  - Not legally compulsory, but **socially important**
  - Ethics guide behavior
  - No legal punishment for breaking ethics
  - Violating ethics leads to **loss of trust and reputation**
- **Example:**
  - Not misusing user data even if it is legal
  - Designing AI that is fair and unbiased

Law

- **Law** is:
  - A **rule made by the government**
  - **Mandatory** for everyone
  - Enforced by courts and legal systems
- **Important points:**
  - Everyone must follow the law
  - Breaking the law leads to **punishment**
  - Punishment may be **fine, jail, or both**
- **Example:**
- Data protection laws (GDPR, DPDP Act)
- Cybercrime laws

Regulation

- Regulation means:
  - Detailed instructions to follow the law
  - Provides procedures and standards
- Example:
  - Traffic law: Follow speed limits
- Regulation:
  - City → 40 km/h
  - Highway → 80 km/h

How Ethics, Law, and Regulation Apply to AI

- In AI systems:
  - **Ethics** → AI should be fair
  - **Law** → AI must not violate human rights
  - **Regulation** → How AI systems should be designed and used
- Order is important:

  Ethics → Law → Regulation

Who Makes AI Laws and Regulations?

- AI laws and regulations are created by:
  - Governments
  - Courts
  - Regulatory bodies
  - International organizations
  - These bodies control how AI is used in society.
- **Examples of Authorities**
- Government
  Creates data protection and privacy laws
- Regulatory bodies
  - Monitor AI usage and compliance
- Courts
  - Decide responsibility if AI causes harm

Why Engineers Must Know Laws and Regulation

- Many students think:
  Laws are only for lawyers
- Reality:
  - AI engineers design systems
  - AI engineers choose data
  - AI engineers decide outputs
- Engineers play a direct role in AI decisions

What Happens If Engineers Ignore Laws?

- If laws are ignored:
  - Company may face legal action
  - Users may be harmed
  - Engineer may be held responsible
- Legal awareness is part of being a professional engineer

Laws and Regulations Governing Artificial Intelligence

1.Data Protection and Privacy Laws

AI systems use **huge amounts of data**, especially **personal data**.

Digital Personal Data Protection Act, 2023(India)

- Protects personal data of individuals
- Data must be collected with consent
- Data must be used only for a valid purpose

If AI misuses personal data, it violates this law.

Example:

Using user photos or phone numbers without permission to train AI → Illegal.

International law

1.GDPR (General Data Protection Regulation – Europe)

- Strongest data protection law
- Gives users rights over their data
- Requires transparency in automated decisions

2. Information Technology and Cyber Laws

These laws control **software systems and digital platforms**.

Information Technology Act, 2000(India)

- Regulates computer systems
- Controls cybercrime
- Ensures data security
- AI software comes under this act.

Example:

If an AI system causes data leakage → IT Act applies.

3. Anti-Discrimination and Equality Laws

- AI must not discriminate unfairly.
- These laws ensure:
  - Equal treatment
  - No bias based on gender, caste, race, religion
  - If AI system discriminates, it violates equality laws.
- **Example:**

AI rejecting resumes only from women → Illegal.

4.Consumer Protection Laws

- AI products are used by customers.
- **Consumer Protection Act, 2019(India)**
  - Protects consumers from unfair practices
  - Companies are responsible for AI-based services
- **Example:**
  AI app gives wrong financial advice →
  Company is responsible.

5.Safety and Liability Laws

- AI systems can cause physical harm.
  Applied to:
  - Autonomous vehicles
  - Medical AI
  - Industrial robots
- Main question:
  Who is responsible if AI causes harm?
- Developer
- Company
- Operator
- Laws decide liability and accountability.

Ethics of the Ethics of AI

- Understanding who defines AI ethics and whether they truly protect society.

Introduction – Ethics of AI Ethics?

- AI is guided by fairness, transparency, and accountability.
- But who decides these principles, and are they themselves ethical?
- Ethics of the Ethics of AI examines how AI ethics are created and whose values they represent.

Understanding AI Ethics

- AI Ethics refers to moral guidelines for AI design and use.
- Core principles:
  - Fairness and non-discrimination
  - Transparency and explainability
  - Accountability
  - Privacy and data protection

Why AI Ethics Alone Is Not Enough

- Despite AI ethics, problems continue:
  - Biased algorithms
  - Mass surveillance
  - Data misuse
  - Lack of accountability

Ethical rules must be questioned and examined.

What Is Ethics of the Ethics of AI?

- It evaluates AI ethical frameworks themselves.

- It questions:
- Who defines ethics?
- Power and bias
- Fairness of ethical rules

Who Defines AI Ethics?

- AI ethics are created by:
- Governments
- Tech companies
- Academic institutions
- International organizations
- Often influenced by powerful corporations and developed nations.

Power and Bias in AI Ethics

- Ethical frameworks may favor:
  - Corporate interests
  - State surveillance
  - Economic profit

- Key question: Are ethics for control or care?

Cultural and Global Challenges

- Ethics are not universal.
- Privacy and surveillance expectations differ across cultures.
- One global ethics model may ignore local values.

Ethics Washing in AI

- **Ethics washing** in AI means a company *pretends* to care about ethical and responsible AI, but doesn’t actually follow those principles in practice.
- Organizations may:
  - Publicly promote ethical AI
    - It is like saying: “We care about fairness, privacy, and transparency.”
  - Privately ignore ethical practices
    - They hide how their AI works.
    - They ignore bias in their systems.
    - They misuse user data.
  - Ethics becomes marketing, not commitment.

- **Example:**
- Imagine a company says their AI hiring tool is "**fair and unbiased**."
  But they never properly test it — and it unfairly rejects certain groups of people.
  They keep advertising it as ethical anyway.
- That's **ethics washing**.

1. Legitimacy

- **AI ethics must be created by legitimate and representative bodies.**
  - AI ethics should not be decided by just one company or a small group of people.
  - Instead, it should be created by **legitimate and representative bodies** — meaning:
    - Proper authorities (like governments)
    - Experts (AI researchers, ethicists, lawyers)
    - Civil society (NGOs, community groups)
    - People who are affected by AI systems
- Because AI affects everyone, the rules should be made by everyone — not just tech companies.

- If only companies decide AI ethics:
  - They may prioritize profit over safety.
  - They may ignore harms to certain communities.
  - They may avoid strict rules.
- But when multiple groups are involved:
  - Different perspectives are considered.
  - People’s rights are protected.
  - Rules become more fair and balanced.

2. Inclusiveness

- **Inclusiveness** means *everyone who is affected gets a voice*.
- When we talk about AI ethics, inclusiveness means the rules guiding AI should not be designed only by powerful companies or wealthy countries.
- They must include people from different backgrounds — especially those who are often ignored.
- If some groups are left out, the ethics themselves become unfair.
- Ethical frameworks must include:
- Marginalized groups
- Developing countries
- Diverse social and cultural voices

1. Marginalized Groups

- These are communities that are often overlooked or disadvantaged, such as:
  - – Minority communities
  - – Low-income populations
  - – People with disabilities
  - – Women and gender minorities
- AI systems can sometimes discriminate against them if their voices are not included in decision-making.

2. Developing Countries

- Many AI systems are created in wealthy nations like the United States or countries in Europe, but they are used worldwide.
- People in:
- Africa
- South Asia
- Latin America
- may have very different social, economic, and cultural realities.
  If they are not included, AI policies may not suit their needs.

- **Diverse Social and Cultural Voices**
- Ethics is not the same everywhere.
- Different cultures may have different views on:
- Privacy,Family,Community &Freedom of speech
- If AI ethics reflects only one culture’s values, it may harm others.

3. Transparency of Ethics

- Ethical rules must be transparent.
- People should know:
- • Who created them
- • Why they were created
- • Whose interests they serve

3. Transparency of Ethics

- When we say **ethical rules must be transparent**, it means the rules should be **clear and open**, not secret or confusing.
- People should clearly know:
- **Who created the rules**
  → Was it a company? The government? Experts? A group of people?
- **Why the rules were created**
  → Are they made to protect people?
  - To make money?
  - To control something?
- **Whose interests the rules protect**
  → Do they protect the public?
  - Only the company?
  - Certain groups?

4. Accountability

- **Accountability** means: *Someone must be responsible when something goes wrong.*
  - Ethics are not just nice words — they must be followed seriously.
- Questions:
- • Who audits ethics?
- • Who is responsible?
- • What happens if ethics are ignored?

- **Who Audits Ethics?**
- An **audit** means checking whether the AI system follows ethical rules.
  - Audits can be done by:
    - Independent third-party experts
    - Government regulators
    - Internal compliance teams (but this alone is not enough)
    - Public oversight bodies

- **Who Is Responsible?**
  - – When AI causes harm, responsibility must be clear:
    - Is it the company that built the AI?
    - The developers who designed it?
    - The organization that deployed it?
    - The government that approved it?
  - – Clear responsibility prevents “blame shifting,” where everyone denies fault.

5. Enforceability

- If ethics are only voluntary, organizations can simply ignore them without any punishment. That makes the rules weak and ineffective.
- For ethical principles to matter, they must:
- **Influence laws** – Governments should turn important ethical principles into actual laws or regulations.
- **Have consequences** – If someone breaks the rules, there should be penalties like fines, legal action, or restrictions.
- **Be enforceable in practice** – There must be systems to monitor, check, and ensure that the rules are truly followed.
- In simple words:
  Ethics should not be optional.They must be **rules that are applied, monitored, and enforced**.

Role of Engineers and Developers

- AI systems are built by engineers and developers. So if something goes wrong, it is not just “the machine’s fault.” It is the responsibility of the people who designed it.
- Engineer must :
- Understand ethical impact
    - They should think:
    - Will this technology harm anyone?
    - Is it unfair to certain people?
    - Can it be misused?
  - Before building something, they must think about its effects on society.

- Question unethical demands
  - Sometimes a manager may say:
    - “Release it quickly.”
    - “Ignore the bias for now.”
    - “Don’t tell users about the risks.”
    - A responsible engineer should not blindly follow such instructions.
- Design fair and transparent systems
  - Technology should:
    - Treat everyone equally
    - Not discriminate
    - Be clear about how it works
- For example:
  - If an AI system selects students for scholarships, it should not unfairly reject students based on gender, caste, race, or background.
  - It should clearly explain how selection is done.
- Ethics is part of professional duty.

- **What is an Artificial Entity?**
- An artificial entity is something created by humans that can:
  - Make decisions
  - Learn from data
  - Interact with people
  - Perform tasks automatically
- For example:
  - Siri answers questions.
  - Alexa controls home devices.
  - ChatGPT helps answer questions and write content.

Introduction – Human Relationship with Artificial Entities

- Artificial entities such as AI systems, robots, chatbots, and virtual assistants are becoming part of everyday human life.
  - AI communicates with humans
  - AI assists in decision-making
  - AI sometimes behaves like a companion
  - This growing interaction creates ethical concerns about emotions, trust, responsibility, and human values.

Why Ethics Matter in Human–AI Relationships

- As AI becomes more human-like:
  - People may emotionally connect with AI
  - AI may influence opinions and decisions
  - Human dignity and social values may be affected
  - Therefore, ethical awareness is essential in human–AI relationships.

Emotional Attachment to Artificial Entities

- When AI talks kindly, remembers your preferences, and responds like a human, people may start feeling attached to it.
- Example:
  - Someone who feels lonely might talk daily with a chatbot like **ChatGPT** and feel emotionally supported.
  - A person may depend heavily on a virtual assistant like **Siri**.
- Even though AI sounds caring, it does not truly feel emotions. But humans may still form emotional bonds.
- **Why this matters:**
  Too much emotional dependence on machines may reduce real human relationships.

Deception and Illusion of Humanity

- Today, some AI systems are designed to act very much like humans. They may have:
  - A human-like voice
  - Facial expressions (in robots or avatars)
  - Emotional-sounding responses
- Because of this, people may forget they are talking to a machine.
- For example:
  - A chatbot on a website pretends to be a customer support agent.
  - A social media account uses AI to chat as if it were a real person.
- This creates what we call **ethical deception** — misleading someone into believing something that is not true.

Trust and Over-Reliance on AI

- Today, many people depend on AI systems for important decisions. Because AI often gives quick and confident answers, people may trust it too much.
- **Examples of Where This Happens**
- **1. Medical AI**
  - Some hospitals use AI to analyze scans and detect diseases.
- If the AI says, “This patient is healthy,” a doctor might rely heavily on that result.
- But what if the AI misses a small tumor?
- **2. Financial Recommendation Systems**
  - Investment apps and banking systems use AI to suggest where to invest money.
- For example, AI systems used by companies like **Robinhood** recommend stocks or financial decisions.
- If the AI makes a wrong prediction, the person may lose money.

Responsibility and Moral Accountability

- As AI systems become more powerful, an important question arises:
- If AI causes harm, who is responsible?
- Imagine these situations:
  - A self-driving car causes an accident.
  - A medical AI gives a wrong diagnosis.
  - An AI hiring system unfairly rejects a candidate.
- Now we must ask:
Is the AI responsible?
Or the developer?
Or the company?
Or the user?

Human Dignity and Respect

- Human dignity means every person deserves respect, kindness, and fair treatment.
- As AI becomes more human-like (talking, responding emotionally, acting polite), people sometimes start treating AI like real people.
- This creates new ethical concerns.
- **Possible Problems**
- **1. Reducing Respect for Real Human Relationships**
- If someone spends more time talking to AI than real people, they may slowly reduce real human interaction.
- For example:
- A person may prefer talking to an AI assistant instead of family or friends because AI is always patient and never disagrees.
- But real relationships require:
- Patience
- Understanding
- Compromise
- AI relationships are easier — but not real.
- Over time, this may weaken social skills and emotional maturity.

6. Dependency and Social Isolation

- As AI becomes more friendly and human-like, some people may start depending on it too much.
- Why?
- Because AI:
- Is always available (24/7)
- Never gets tired
- Never argues
- Often agrees with you
- Does not judge you
- This can make AI interaction feel easier than real human relationships.

Manipulation and Behavioral Control

- AI systems today are very powerful. They don’t just give information — they can also influence:
- Emotions
- Opinions
- Decisions
- Sometimes this influence is helpful. But sometimes it can become manipulation.

Privacy in Human–AI Relationships

- When people interact with AI systems, they often share very personal information — sometimes without realizing it.
- AI entities can collect:
- Conversations (what you say or type)
- Emotions (how you feel, based on your words or tone)
- Behavior patterns (what you search, buy, watch, or click)
- This For example:
- A chatbot like **ChatGPT** processes your questions and responses.
- A voice assistant like **Siri** listens for voice commands.
- Platforms like **Google** collect browsing and search data to personalize services.
- The more you use AI, the more data it gathers about you.
- creates serious privacy concerns.

Conclusion

- “Our relationship with artificial entities is not just a technical issue — it is a moral issue.
- AI may not have feelings, but humans do. How we design and interact with AI affects:
- Human dignity
- Social relationships
- Moral responsibility
- Therefore, ethical awareness in human–AI relationships is essential for a healthy society.”